

Statistics, Counting & Probability

ANSWER KEY



The fundamental counting principle

- 1 A restaurant menu has 4 appetizers, 6 main courses, and 3 desserts. How many different 3-course meals are possible?

By the fundamental counting principle: $4 \times 6 \times 3 = 72$.

- 2 A license plate consists of 3 letters followed by 4 digits. How many different license plates are possible, assuming repetition is allowed?

$$26^3 \times 10^4 = 17576 \times 10000 = 175,760,000.$$

Permutations

- 3 In how many ways can 5 different books be arranged on a shelf?

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120.$$

- 4 In how many ways can a president, vice president, and secretary be chosen from a club of 10 members?

$$P(10, 3) = \frac{10!}{7!} = 10 \times 9 \times 8 = 720.$$

Combinations

- 5 In how many ways can a hand of 5 cards be dealt from a standard 52-card deck?

$$\binom{52}{5} = \frac{52!}{5!47!} = 2,598,960.$$

- 6 A committee of 4 is chosen from 6 men and 5 women. In how many ways can the committee consist of exactly 2 men and 2 women?

$$\binom{6}{2} \times \binom{5}{2} = 15 \times 10 = 150.$$

Probability with combinations

- 7 A bag contains 5 red and 7 blue marbles. If 3 marbles are drawn at random without replacement, find the probability all 3 are red

$$P(\text{all red}) = \frac{\binom{5}{3}}{\binom{12}{3}} = \frac{10}{220} = \frac{1}{22}.$$

The binomial probability formula

- 8 A fair coin is flipped 5 times. Find the probability of getting exactly 3 heads, using the binomial probability formula $P(k) = \binom{n}{k} p^k (1-p)^{n-k}$.
- $$P(3) = \binom{5}{3} (0.5)^3 (0.5)^2 = 10(0.125)(0.25) = 10(0.03125) = 0.3125.$$
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Expected value and variance

- 9 A game gives a payout of 10 dollars with probability 0.3, 5 dollars with probability 0.4, and 0 dollars with probability 0.3. Find the expected payout.
- $$\text{Expected value} = 10(0.3) + 5(0.4) + 0(0.3) = 3 + 2 + 0 = 5 \text{ dollars.}$$
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Normal distribution and z-scores

- 10 A data set is normally distributed with mean 70 and standard deviation 8. Find the z-score for a data value of 86.
- $$z = \frac{86 - 70}{8} = \frac{16}{8} = 2.$$
- 11 Using the empirical rule (68-95-99.7 rule), estimate the percentage of data within 2 standard deviations of the mean.
- By the empirical rule, approximately 95% of data in a normal distribution falls within 2 standard deviations of the mean.
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A multi-part probability and counting problem

- 12 This question has two parts. A class has 8 boys and 7 girls. A group of 4 students is selected at random to represent the class. (a) Find the total number of ways to select the group of 4 from the 15 students. (b) Find the probability that the group consists of exactly 2 boys and 2 girls.
- $$(a) \binom{15}{4} = \frac{15!}{4!11!} = 1365. (b) \text{ Ways to choose 2 boys and 2 girls: } \binom{8}{2} \times \binom{7}{2} = 28 \times 21 = 588.$$
- $$\text{Probability: } \frac{588}{1365} \approx 0.431, \text{ or about } 43.1\%.$$
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Permutations with repeated elements

- 13 Find the number of distinct arrangements of the letters in the word "BANANA".
- The word has 6 letters with repeats: A appears 3 times, N appears 2 times. Number of distinct arrangements: $\frac{6!}{3!2!} = \frac{720}{12} = 60.$
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Conditional probability with counting

- 14** A committee of 3 is selected from 5 men and 4 women. Given that at least one woman is selected, find the probability that exactly 2 women are selected.

Total ways with at least 1 woman: total ways $\binom{9}{3} = 84$ minus all-men ways $\binom{5}{3} = 10$, giving $84 - 10 = 74$. Ways with exactly 2 women: $\binom{4}{2} \times \binom{5}{1} = 6 \times 5 = 30$. Conditional probability: $\frac{30}{74} = \frac{15}{37} \approx 0.405$.

Percentiles and standardized scores

- 15** A student scores at the 85th percentile on a standardized test. Explain what this means in terms of the rest of the test-taking population.

Scoring at the 85th percentile means the student performed better than approximately 85% of all test-takers, and approximately 15% of test-takers scored higher than this student.

Mutually exclusive vs. overlapping events

- 16** A die is rolled once. Find $P(\text{rolling an even number or a 5})$, explaining why the addition rule simplifies since these events are mutually exclusive.

Even numbers: 2, 4, 6 (3 outcomes). Since 5 is odd, it cannot overlap with the even outcomes — the events are mutually exclusive. $P(\text{even or } 5) = P(\text{even}) + P(5) = \frac{3}{6} + \frac{1}{6} = \frac{4}{6} = \frac{2}{3}$.

A multi-part counting and probability problem with restrictions

- 17** This question has three parts. A password consists of 4 characters chosen from the 10 digits 0-9, and repetition is not allowed. (a) Find the total number of possible passwords. (b) Find the number of passwords where the first character is not 0 (a common restriction to avoid leading zeros). (c) Find the probability that a randomly generated valid password (from part b) starts with an even digit.

(a) $P(10, 4) = 10 \times 9 \times 8 \times 7 = 5040$. (b) First character has 9 choices (1-9, excluding 0); remaining 3 characters chosen from the remaining 9 digits without repetition: $9 \times 9 \times 8 \times 7 = 4536$. (c) Even first digits from 1-9: 2, 4, 6, 8 (4 choices). Passwords starting with an even digit: $4 \times 9 \times 8 \times 7 = 2016$. Probability: $\frac{2016}{4536} = \frac{4}{9} \approx 0.444$.